Zero Shot Sentiment Analysis on Tweets in Any Language

Oscar I Hernandez* (ohernandez13@simons-rock.edu), Mike Izbicki (mike@izbicki.me)

ARKS*, Claremont McKenna College

Objective

Given sentiment-labeled tweets in a popular language, we train a convolutional neural network which transfer learns to evaluate sentiment in any other language with unlabeled tweets.

Introduction

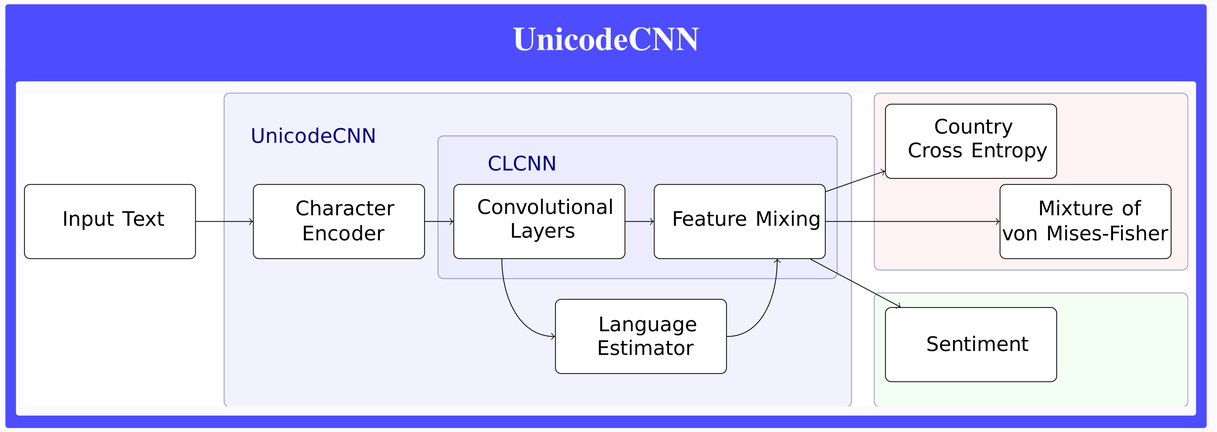
We repurpose the UnicodeCNN [1] – a recent state-of-the-art model which geolocates tweets in any language at any location by analyzing text in Unicode – to analyze sentiment by replacing the final layer and pretraining the rest. Our model offers two advantages over existing sentiment classifiers:

1. It is pretrained on tweet geolocation, which grows massive amounts of reliable data.
2. It is already capable of analyzing text in any language by looking at its underlying subcharacter representation in Unicode.

The example tweets below exemplify how similarity in location and Unicode can inform sentiment across languages.

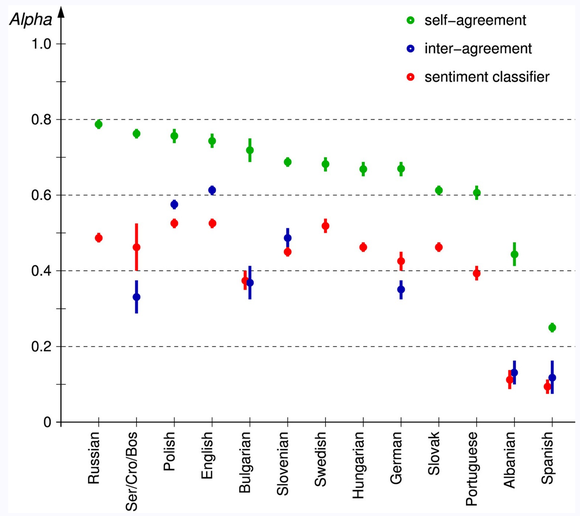


UnicodeCNN



Dataset

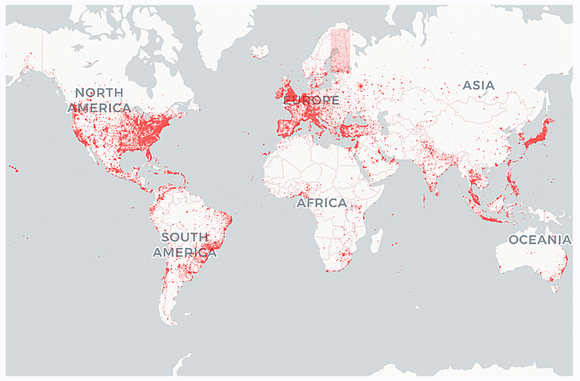
We train using a publicly available sentiment-labeled dataset shown below [2] containing 1.6 million tweets in 13 languages, which is imprecise for some languages due to the difficulty required to label at this scale. Because English comprises for 42% of all tweets, we aim to use its popularity to learn to classify sentiment for the *low resource languages* which lack sentiment-labeled datasets.



We also leverage the UnicodeCNN's dataset of all 900 geo-encoded tweets sent over 9 months by over 3.0 million users in over 100 languages.

Model

The UnicodeCNN [1] predicts the exact GPS coordinate of a tweet with an output layer based on the mixture of von Mises-Fisher distribution and multitask learns the country cross entropy to accelerate training. As shown above, we replace the output layers with a single layer of sentiment classification.



In each experiment, we do the following.

1. Start with the model pre-trained to geolocate the 900 million tweets whose locations are shown above.
2. Choose to train all layers or just final layers.
3. Train on sentiment-labeled tweets.
4. Test separately on unlabeled tweets in either language.

Contribution

In the main experiments 3 and 6 highlighted in the table below, we train on sentiment-labeled English tweets and test on unlabeled Portuguese tweets. In two separate experiments 2 and 5, we determine the accuracy of our model by training models on sentiment-labeled Portuguese tweets.

#	Retrain	Train	Test
1	All	English	English
2	All	Portuguese	Portuguese
3	All	English	Portuguese
4	Final	English	English
5	Final	Portuguese	Portuguese
6	Final	English	Portuguese

We chose English as the source *high resource language* to build a model with lots of high-quality data. We chose to target Portuguese because, although we do not need sentiment-labeled tweets in our target language to train our model, we do need to train a new model on its sentiment-labeled dataset in order to validate the results.

After finalizing the results on Portuguese, we aim to test and validate our English-trained model on other languages. If successful, we would expect it to succeed on other *low resource languages* for which sentiment-labeled datasets do not exist.

References

[1] Mike Izbicki, Evangelos Papalexakis, and Vassilis Tsotras.
Geolocating tweets in any language at any location.
CIKM, 2019.

[2] Igor Mozetič, Miha Grčar, and Jasmina Smailović.
Multilingual twitter sentiment classification: The role of human annotators.
PLOS ONE, 11(5):1–26, 05 2016.

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